

Appendix

Hyperparameter Selection

Table 1: Hyperparameter search spaces for all tunable models. All time series models are selected via exhaustive grid search on the validation set; the trace-level models (ProcessTransformer, GenerativeLSTM) use Bayesian optimization. Fixed parameters not listed here are held constant across all candidates.

Model	Hyperparameter	Search Space
ETS	Error component	{add, mul}
	Trend component	{none, add, mul}
	Seasonal component	{none, add, mul}
	Damped trend	{True, False}
SARIMAX	ARIMA order (p, d, q)	{{(1,0,0),(1,1,0),(0,1,1),(1,1,1)}}
	Seasonal order (P, D, Q, s)	{{(0,0,0,0),(1,0,0,7),(0,1,1,7),(1,1,1,7)}}
Theta	θ	{1.5, 2.0, 3.0}
	Deseasonalise period	{7, 14, none}
Ridge / Ridge _M	Lags (lookback)	{7, 14, 28}
	Regularisation α	{0.01, 0.1, 1, 10}
N-HITS	Input chunk length	{14, 28}
	Output chunk length	{7, 14}
	Layer width	{256, 512}
TFT	Input chunk length	{14, 28}
	Output chunk length	{7, 14}
	Hidden size	{16, 32}
TabPFN	Lookback n_{steps}	{7, 14, 28}
PT	Attention heads	{2, 4}
	Batch size	{16, 32}
	Epochs	{20, 50}
	Learning rate	{ 5×10^{-4} , 10^{-3} , 5×10^{-3} }
GLSTM	Model type	{shared_cat, concatenated}
	LSTM activation	{selu, tanh}
	Learning rate	{ 5×10^{-4} , 10^{-3} , 2×10^{-3} }
	Dropout	{0.1, 0.2, 0.3}

Table 2: Average MAE $\pm$ standard deviation in CTP across three random seeds in LoAp logs.

		LoAp (simple)					
		Flat	Season.	Trend	Drift	Comb.	
Basel.	Naïve	14.13 ± 0.00	15.38 ± 0.00	45.93 ± 0.00	15.71 ± 0.00	187.47 ± 0.00	
	Val. Avg.	27.71 ± 0.00	29.65 ± 0.00	109.32 ± 0.00	23.99 ± 0.00	254.37 ± 0.00	
Statistical	ETS	14.13 ± 0.00	10.97 ± 0.00	345.35 ± 0.00	49.73 ± 0.00	247.01 ± 0.00	
	SARIMAX	75.81 ± 0.00	101.08 ± 0.00	44.94 ± 0.00	99.24 ± 0.00	12.36 ± 0.00	
	STL	14.12 ± 0.00	13.75 ± 0.00	48.28 ± 0.00	12.96 ± 0.00	45.30 ± 0.00	
	Prophet	645.10 ± 0.00	561.89 ± 0.00	121.06 ± 0.00	173.59 ± 0.00	23.37 ± 0.00	
ML	Ridge	12.43 ± 0.00	15.26 ± 0.00	17.68 ± 0.00	23.99 ± 0.00	85.28 ± 0.00	
	Ridge _M	18.50 ± 0.00	21.87 ± 0.00	19.62 ± 0.00	23.49 ± 0.00	96.44 ± 0.00	
	N-BEATS	20.14 ± 2.22	19.26 ± 0.54	42.03 ± 13.68	28.51 ± 5.60	110.41 ± 88.58	
	TFT	29.05 ± 1.86	32.91 ± 2.27	180.58 ± 84.44	52.07 ± 6.92	292.52 ± 141.89	
FM	Chronos TabPFN						
SIM	PMSD	19.75 ± 0.00	12.83 ± 0.00	62.91 ± 0.00	22.08 ± 0.00	79.70 ± 0.00	
	Simod AgentSimulator						
Trace-Level PPM	Plain	GLSTM	162.46 ± 37.95	243.76 ± 18.57	205.67 ± 39.94	158.27 ± 20.93	615.29 ± 50.34
		PGT	133.21 ± 2.65	91.30 ± 1.81	167.18 ± 1.56	88.49 ± 2.25	361.27 ± 3.40
		PT _{RT}	106.10 ± 19.34	168.92 ± 0.49	162.25 ± 44.02	149.34 ± 13.10	568.06 ± 146.47
		PT _{su}	74.17 ± 19.71	116.99 ± 13.05	76.47 ± 41.57	76.34 ± 8.85	413.50 ± 102.49
	First	GLSTM	224.71 ± 29.78	240.72 ± 30.61	301.21 ± 30.34	167.39 ± 15.22	659.89 ± 48.05
		PF Amiri	62.68 ± 2.65	62.23 ± 2.00	91.33 ± 1.54	38.38 ± 2.25	430.12 ± 3.40
		PT _{su}	105.91 ± 60.22	130.76 ± 12.84	181.27 ± 45.78	90.64 ± 7.67	467.94 ± 94.19
		PT _{RT}	158.77 ± 55.91	182.06 ± 0.50	228.01 ± 104.41	162.86 ± 12.90	617.57 ± 134.33
	Half	GLSTM	121.97 ± 19.89	127.22 ± 13.18	165.02 ± 16.98	89.00 ± 11.23	383.05 ± 36.37
		Amiri	112.82 ± 8.72	108.38 ± 6.20	181.87 ± 36.60	89.98 ± 9.03	357.04 ± 7.06
		PT _{su}	114.72 ± 42.72	92.13 ± 6.30	137.14 ± 23.64	114.30 ± 3.42	78.10 ± 41.27
		PT _{RT}	104.50 ± 27.63	125.67 ± 5.81	155.36 ± 48.30	92.95 ± 2.45	381.65 ± 88.13

Robustness

We here report the average MAE $\pm$ standard deviation in CTP across three random seeds for all the models, indicating their robustness to random weight initialization. We show only CTP in the synthetic logs, the rest can be found in the file `robustness_by_dataset.xlsx` in the `/results` folder. Note that many approaches are deterministic whereas some ML-based TS models show fluctuations and trace-level PPM approaches vary even more.

Table 3: Average MAE $\pm$ standard deviation in CTP across three random seeds in O2C logs.

		O2C					
		Flat	Season.	Trend	Drift	Comb.	
Basel.	Naïve	26.43 ± 0.00	80.89 ± 0.00	485.04 ± 0.00	48.53 ± 0.00	1021.45 ± 0.00	
	Val. Avg.	22.55 ± 0.00	48.64 ± 0.00	596.61 ± 0.00	47.06 ± 0.00	1337.14 ± 0.00	
Statistical	ETS	21.31 ± 0.00	40.69 ± 0.00	171.14 ± 0.00	32.94 ± 0.00	172.44 ± 0.00	
	SARIMAX	73.03 ± 0.00	41.21 ± 0.00	71.62 ± 0.00	67.69 ± 0.00	105.79 ± 0.00	
	STL	24.76 ± 0.00	22.26 ± 0.00	159.55 ± 0.00	32.59 ± 0.00	100.02 ± 0.00	
	Prophet	1498.57 ± 0.00	1536.06 ± 0.00	1747.54 ± 0.00	395.31 ± 0.00	637.37 ± 0.00	
ML	Ridge	67.84 ± 0.00	124.77 ± 0.00	114.98 ± 0.00	128.07 ± 0.00	236.02 ± 0.00	
	Ridge _M	53.89 ± 0.00	54.68 ± 0.00	1100.86 ± 0.00	22.85 ± 0.00	1334.02 ± 0.00	
	N-BEATS	41.57 ± 7.14	58.46 ± 4.27	688.26 ± 271.25	42.58 ± 1.91	236.24 ± 103.67	
	TFT	101.00 ± 47.03	108.87 ± 58.36	847.55 ± 306.49	107.82 ± 39.20	1436.94 ± 709.97	
FM	Chronos TabPFN						
SIM	PMSD	47.24 ± 0.00	70.16 ± 0.00	263.24 ± 0.00	32.72 ± 0.00	70.97 ± 0.00	
	Simod AgentSimulator						
Trace-Level PPM	Plain	GLSTM	1194.79 ± 43.59	1185.21 ± 61.95	1457.74 ± 156.73	1139.23 ± 34.41	3056.05 ± 58.62
		PGT	466.13 ± 10.64	1069.96 ± 19.21	1958.61 ± 17.02	1052.34 ± 19.42	3262.59 ± 61.64
		PT _{RT}	776.67 ± 439.73	1574.14 ± 15.80	2315.54 ± 14.07	1445.93 ± 31.68	3604.15 ± 398.87
		PT _{su}	603.40 ± 194.59	1208.09 ± 56.40	1981.31 ± 33.65	1157.70 ± 62.86	2959.47 ± 223.89
	First	GLSTM	1032.27 ± 44.90	1553.39 ± 99.20	2318.13 ± 48.69	1466.30 ± 22.09	3793.55 ± 22.75
		PGT	405.28 ± 10.64	398.52 ± 19.21	432.82 ± 17.02	393.45 ± 19.42	1792.29 ± 61.64
		PT _{su} full	782.70 ± 171.69	850.94 ± 88.36	1204.94 ± 75.61	831.86 ± 97.43	2053.87 ± 422.49
		PT _{RT} full	935.45 ± 452.93	1257.72 ± 28.01	1542.26 ± 38.72	1087.59 ± 62.23	2713.15 ± 749.25
	Half	GLSTM	746.22 ± 44.03	688.11 ± 67.67	799.05 ± 129.59	667.06 ± 25.08	2225.76 ± 86.94
		Amiri	454.02 ± 32.89	446.84 ± 160.43	656.11 ± 65.18	489.40 ± 82.85	1604.85 ± 261.43
		PT _{su}	214.18 ± 111.22	270.64 ± 67.12	362.62 ± 67.58	209.48 ± 76.68	1021.73 ± 301.95
		PT _{RT}	584.76 ± 384.99	837.21 ± 17.13	1009.88 ± 39.91	694.59 ± 1.77	1910.66 ± 736.50